Complex Number 20

1. The point P represents the complex numbers z on an Argand diagram. Describe the locus of P in each of the following cases and find the complex number z which satisfies both of these equations
   a. $|z| = |z-2|$
   b. $\arg(z-2) = \arg(z+2) + \frac{\pi}{2}$
2. Show that if $-1 < r < 1$, then $1 + r\cos\theta + r^2\cos 2\theta + r^3\cos 3\theta + \cdots = \frac{1-r\cos\theta}{1-2r\cos\theta+r^2}$
3. Let w be a seventh root of unity. Find the equation of the quadratic polynomial with roots $w + w^2 + w^4$ and $w^3 + w^5 + w^6$.

4. a. If $z = \cos\theta + i\sin\theta$, prove that $z^n + \frac{1}{z^n} = 2\cos n\theta$.
   b. Solve $z^8 + 1 = 0$ and use your solutions to factorise $z^8 + 1$ into real quadratic factors.
   c. Use your answer to part **a** and **b** to prove $\cos 4\theta = 8\left(\cos^2\theta - \cos^2\frac{\pi}{8}\right)\left(\cos^2\theta - \cos^2\frac{3\pi}{8}\right)$.

5. a. Given $w = cis\frac{2\pi}{9}$, show that $1 + w + w^2 + w^3 + w^4 + w^5 + w^6 + w^7 + w^8 = 0$.
   b. Hence or otherwise show that $(w^4 + w^5)(w^2 + w^7) + (w^3 + w^6)(w^2 + w^7) = -1$.
   c. Hence or otherwise show that $\cos\frac{4\pi}{9}\left(2\cos\frac{8\pi}{9} - 1\right) = -\frac{1}{2}$.

6. z_1 and z_2 are two complex numbers such that $|z_1| = |z_2|$ and $0 < \arg z_1 < \arg z_2 < \frac{\pi}{2}$. On an Argand diagram vectors OP and OQ represent z_1 and z_2 respectively. $Q\hat{O}P = \frac{\pi}{3}$ and $OPRQ$ is a parallelogram.
   a. Draw a diagram to show the above information.
   b. Find the value of $\arg\left(\frac{z_2}{z_1}\right)$
   c. Show that $z_1 + z_2 = \sqrt{3}i(z_1 - z_2)$

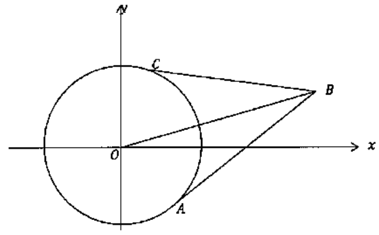
7. Given that w is a complex cube root of unity,
   a. Prove that $1 + w + w^2 = 0$.
   b. Prove that $1 + w^2 = \frac{1}{1+w}$.
   c. Prove that $1 + w$ is a complex cubic root of -1.
   d. Express the other complex cubic root of -1 as a quadratic expression in w.

8. If $|z| = 1$, prove that $\frac{1+z}{1+\bar{z}} = z$.
9. If $z = \cos\theta + i\sin\theta$, prove that $\frac{2}{1+z^2} = 1 - i\tan\theta$.
10. Let $z = \cos\theta + i\sin\theta$
    a. Use de Moivre's theorem to show that $z^n + z^{-n} = 2\cos n\theta$
    b. Show that $(z + z^{-1})^4 = (z^4 + z^{-4}) + 6 + 4(z^2 + z^{-2})$
    c. Hence show that $\cos^4\theta = \frac{1}{8}\cos 4\theta + \frac{1}{2}\cos 2\theta + \frac{3}{8}$
11. Use De Moivre's Theorem to show that: $\sin^4\theta = \frac{1}{8}\cos 4\theta - \frac{1}{2}\cos 2\theta + \frac{3}{8}$. (Start by showing that $z^n - z^{-n} = 2i\sin n\theta$)
12. a. Use De Moivre's Theorem to show: $\cos 6\alpha = 32\cos^6\alpha - 48\cos^4\alpha + 18\cos^2\alpha - 1$
    b. Hence show that the polynomial equation $32x^6 - 48x^4 + 18x^2 - 1 = 0$ has roots of the form $x = \cos\frac{(2n+1)\pi}{12}$
    c. Use the product of these six roots to deduce that $\cos\frac{\pi}{12}\cos\frac{5\pi}{12} = \frac{1}{4}$
13. a. Use De Moivre's Theorem to show that $\cos^5\theta = \frac{1}{16}(\cos 5\theta + 5\cos 3\theta + 10\cos\theta)$
    b. Hence evaluate $\int_0^{\frac{\pi}{2}} \cos^5\theta \; d\theta$

14. Prove that $\left(\frac{1+\cos\theta+i\sin\theta}{1+\cos\theta-i\sin\theta}\right)^n = \cos n\theta + i\sin n\theta$.

15. Let $x = \cos\frac{2\pi}{5} + i\sin\frac{2\pi}{5}$ and $b = x + \frac{1}{x}$, $b > 0$.
   a. Show that $x^4 + x^3 + x^2 + x + 1 = 0$.
   b. Use part a. to show that $b^2 + b - 1 = 0$.
   c. Show that $x^2 - bx + 1 = 0$ and hence show that $x = \frac{1}{2}\left(b + i\sqrt{4-b^2}\right)$
   d. Hence use part b. to show that $\cos\frac{2\pi}{5} = \frac{-1+\sqrt{5}}{4}$ and $\sin\frac{2\pi}{5} = \frac{1}{2}\sqrt{\frac{5+\sqrt{5}}{2}}$.

16. In the diagram below, $\overrightarrow{OA} = z_1, \overrightarrow{OB} = z_2$ and $\overrightarrow{OC} = z_3$. AB and AC are tangents to the circle AC, centre O.
   a. State why $\frac{z_3 - z_2}{z_3}$ and $\frac{z_2 - z_1}{z_1}$ are entirely imaginary.
   b. State why $|z_3 - z_2| = |z_2 - z_1|$
   c. Prove $\frac{z_3 - z_1}{z_2}$ is entirely imaginary.
   d. Find, with reasons, the centre of the circle OABC.



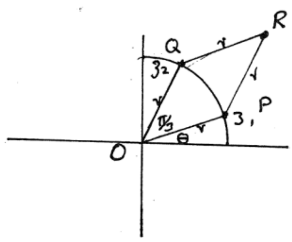
17. By considering that $\cos\theta + i\sin\theta = \cos\theta(1 + i\tan\theta)$ and de Moivres' Theorem.
   a. Find the expression for $\cos 4\theta$ in terms of $\cos\theta$ and $\tan\theta$.
   b. As $\sin 4\theta = \cos^4\theta(4\tan\theta - 4\tan^3\theta)$, find the result for $\tan 4\theta$ in terms of $\tan\theta$.
   c. Show that the solutions to the polynomial equation: $x^4 + 4\sqrt{3}x^3 - 6x^2 - 4\sqrt{3}x + 1 = 0$ can be calculated from $\tan 4\theta = \frac{1}{\sqrt{3}}$.
   d. Find the four solutions to this quartic equation
   e. Hence show that $\tan\frac{7\pi}{24}\tan\frac{11\pi}{24} = \cot\frac{\pi}{24} = \cot\frac{5\pi}{24}$.

18. a. Express i and $(1+i)^n$ in mod-arg form
   b. Simplify $cis\,\theta + cis\,(-\theta)$
   c. Let M, N be positive integers. The polynomial $x^M(1-x)^N$ when divided by $1 + x^2$ has a remainder of $ax + b$. Using the results from a and b above, or otherwise, show that $a = \left(\sqrt{2}\right)^N \sin\frac{(2M-N)\pi}{4}$ and $b = \left(\sqrt{2}\right)^N \cos(\frac{(2M-N)\pi}{4})$

19. a. Show that $\cos 6\theta = 32\cos^6\theta - 48\cos^4\theta + 18\cos^2\theta - 1$.
   b. Hence, find all the roots of the polynomial $32x^6 - 48x^4 + 18x^2 - 1 = 0$.
   c. Show that $\cos\left(\frac{\pi}{12}\right)\cos\left(\frac{5\pi}{12}\right) = \frac{1}{4}$.
   d. Find the exact value of. $\cos^2\left(\frac{\pi}{12}\right) + \cos^2\left(\frac{5\pi}{12}\right)$.
   e. Hence, show that $\cos\left(\frac{\pi}{12}\right) + \cos\left(\frac{5\pi}{12}\right) = \frac{\sqrt{6}}{2}$

Answer

1. $x = 1$ and $y = \sqrt{4 - x^2}$; $1 + \sqrt{3}i$
2. prove
3. $x^2 + x + 2 = 0$
4. b. $z = cis\left(\frac{\pi + 2k\pi}{8}\right)$, $k = 0, 1, \ldots, 7$.
   $z^8 + 1 = \left(z^2 - 2\cos\frac{\pi}{8}z + 1\right)\left(z^2 - 2\cos\frac{3\pi}{8}z + 1\right)\left(z^2 - 2\cos\frac{5\pi}{8}z + 1\right)\left(z^2 - 2\cos\frac{7\pi}{8}z + 1\right)$



6. a.
   b. $\arg\left(\frac{z_2}{z_1}\right) = \frac{\pi}{3}$
7. a. $1 + w^2$

16. b. $\frac{8}{15}$ d. AS $\angle OAB = \angle OCB = \frac{\pi}{2}$, OABC is a cyclic quadrilateral and OB is circle diameter , midpoint of OB is centre of circle OABC.
17. a. $\cos 4\theta = \cos^4\theta(1 - 6\tan^2\theta + \tan^4\theta)$; b. $\tan 4\theta = \frac{4\tan\theta - 4\tan^3\theta}{1 - 6\tan^2\theta + \tan^4\theta}$; c. proving; d. $\tan\frac{\pi}{24}, \tan\frac{7\pi}{24}, -\tan\frac{\pi}{24}, -\tan\frac{5\pi}{24}$
18. a. $\sqrt{2}cis\left(\frac{n\pi}{4}\right)$ b. $2\cos\theta$